GSE544 homework 5 part 1

Bishwa R. Rai with Claude Sonnet 4.6

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*LLM disclaimer: Claude Sonnet 4.6 was used to assist in writing code and to check solutions.

Fitting marginals and copulas to the VMLS portfolio

```
library(tidyverse)
library(MASS)       # fitdistr
library(mvtnorm)    # rmvnorm, dmvnorm
library(copula)     # tCopula, normalCopula, fitCopula, rCopula
```

```
returns <- read_csv("C:/Users/Bishw/Desktop/DA/vmls_portfolio_returns.csv")
```

```
## Rows: 2500 Columns: 20
## -- Column specification --------------------------------------------------------------------------------
## Delimiter: ","
## dbl (20): stock_1, stock_2, stock_3, stock_4, stock_5, stock_6, stock_7, stock_8, stock_9, stock_10,
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Training set: first 2000 rows, risky assets 1-19 (first 19 of 20 columns)
train <- returns[1:2000, 1:19]
colnames(train) <- paste0("asset_", 1:19)

# Four-asset subset: stocks 1, 2, 3, 10 (0-indexed [0,1,2,9] => columns 1,2,3,10)
four_assets <- train[, c(1, 2, 3, 10)]
colnames(four_assets) <- c("stock_1", "stock_2", "stock_3", "stock_10")
```

a) Student-t marginals

```
# Fit Student-t to each of the 19 assets
# fitdistr returns names like "df.df", "loc.m", "scale.s" - store raw and index by grepl
fit_t <- function(x) fitdistr(x, "t")$estimate

marginal_params <- lapply(train, fit_t)

# Build summary table
marginal_params_df <- do.call(rbind, lapply(marginal_params, function(p) {
  c(df = unname(p["df"]), loc = unname(p["m"]), scale = unname(p["s"]))
})) %>%
  as_tibble(rownames = "asset")
colnames(marginal_params_df) <- c("asset", "df", "loc", "scale")

# Print summary for stocks 1, 2, 3, 10
```

```r
marginal_params_df %>%
  filter(asset %in% c("asset_1", "asset_2", "asset_3", "asset_10")) %>%
  knitr::kable(digits = 4, caption = "Student-t marginal fits for stocks 1, 2, 3, 10")
```

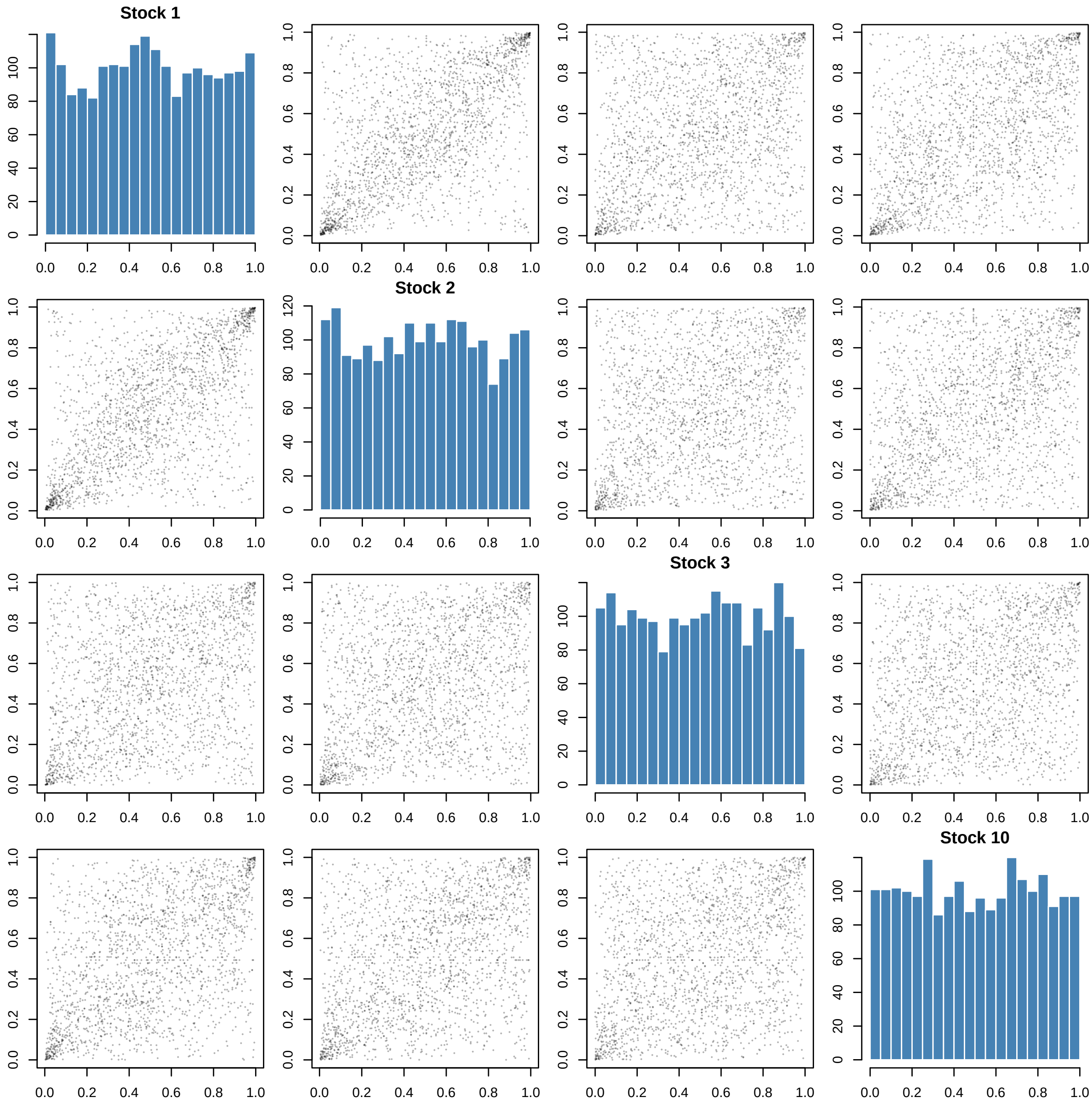
Table 1: Student-t marginal fits for stocks 1, 2, 3, 10

asset	df	loc	scale
asset_1	1.7858	4e-04	0.0115
asset_2	1.3511	-2e-04	0.0106
asset_3	3.0861	8e-04	0.0122
asset_10	4.0148	3e-04	0.0145

All four assets have $\hat{\nu} < 5$, confirming very fat tails. Stock 1 is the fattest-tailed (lowest $\hat{\nu} = 1.79$) and stock 10 is the lightest-tailed ($\hat{\nu} = 4.01$) among the four.

b) PITs and the empirical scatter matrix

```r
# Apply PIT u = F(r) using fitted Student-t marginals, clip to [1e-6, 1-1e-6]
# fitdistr returns names "m", "s", "df" - use directly
u_all <- matrix(NA_real_, nrow = nrow(train), ncol = 19)
for (i in seq_along(marginal_params)) {
  p         <- marginal_params[[i]]
  raw       <- as.numeric(train[[i]])
  u         <- pt((raw - p["m"]) / p["s"], df = p["df"])
  u_all[, i] <- pmin(pmax(u, 1e-6), 1 - 1e-6)
}
u_all <- as.data.frame(u_all)
colnames(u_all) <- paste0("asset_", 1:19)

# Four-asset PIT subset
u4 <- u_all[, c(1, 2, 3, 10)]
colnames(u4) <- c("stock_1", "stock_2", "stock_3", "stock_10")

# 4x4 empirical PIT scatter matrix
# Diagonal: marginal uniformity histograms; off-diagonal: pairwise scatters
labels <- c("Stock 1", "Stock 2", "Stock 3", "Stock 10")

par(mfrow = c(4, 4), mar = c(2, 2, 1.5, 1))

for (i in 1:4) {
  for (j in 1:4) {
    if (i == j) {
      hist(u4[[i]], breaks = 20, col = "steelblue", border = "white",
           main = labels[i], xlab = "", ylab = "")
    } else {
      plot(u4[[j]], u4[[i]], pch = 16, cex = 0.3, col = rgb(0, 0, 0, 0.3),
           xlab = labels[j], ylab = labels[i], main = "")
    }
  }
}
```



The diagonal histograms are approximately uniform, confirming the Student-t marginals are well-fitted. In the off-diagonal scatters, dependence concentrates in the corners, particularly the lower-left (joint lower tail) and upper-right (joint upper tail), indicating co-movement during extreme market events. This corner concentration is the key empirical pattern that any fitted copula must reproduce.

c) Gaussian copula

```r
# Fit Gaussian copula on the four-asset subset
# The correlation matrix is the entire dependence structure
u4_mat <- as.matrix(u4)

gauss_cop <- normalCopula(dim = 4, dispstr = "un")
gauss_fit  <- fitCopula(gauss_cop, data = u4_mat, method = "ml")
```

```r
gauss_cor <- coef(gauss_fit)

# Reconstruct the 4x4 correlation matrix
P_gauss <- diag(4)
idx <- which(lower.tri(P_gauss), arr.ind = TRUE)
for (k in seq_len(nrow(idx))) {
  P_gauss[idx[k, 1], idx[k, 2]] <- gauss_cor[k]
  P_gauss[idx[k, 2], idx[k, 1]] <- gauss_cor[k]
}

rownames(P_gauss) <- colnames(P_gauss) <- labels
P_gauss %>%
  knitr::kable(digits = 4, caption = "Gaussian copula correlation matrix")
```

Table 2: Gaussian copula correlation matrix

	Stock 1	Stock 2	Stock 3	Stock 10
Stock 1	1.0000	0.6613	0.4188	0.5370
Stock 2	0.6613	1.0000	0.3938	0.4687
Stock 3	0.4188	0.3938	1.0000	0.4080
Stock 10	0.5370	0.4687	0.4080	1.0000

d) t-copula

```r
# Fit t-copula on the four-asset subset
t_cop <- tCopula(dim = 4, dispstr = "un")
t_fit  <- fitCopula(t_cop, data = u4_mat, method = "ml")

t_params <- coef(t_fit)
nu_hat   <- t_params["df"]
cat("Fitted copula degrees of freedom:", round(nu_hat, 4), "\n")
```

```
## Fitted copula degrees of freedom: 3.7628
```

```r
# Compare to marginal dfs
marginal_nu <- marginal_params_df %>%
  filter(asset %in% c("asset_1", "asset_2", "asset_3", "asset_10")) %>%
  dplyr::select(asset, df)
marginal_nu %>%
  knitr::kable(digits = 4, caption = "Marginal Student-t degrees of freedom")
```

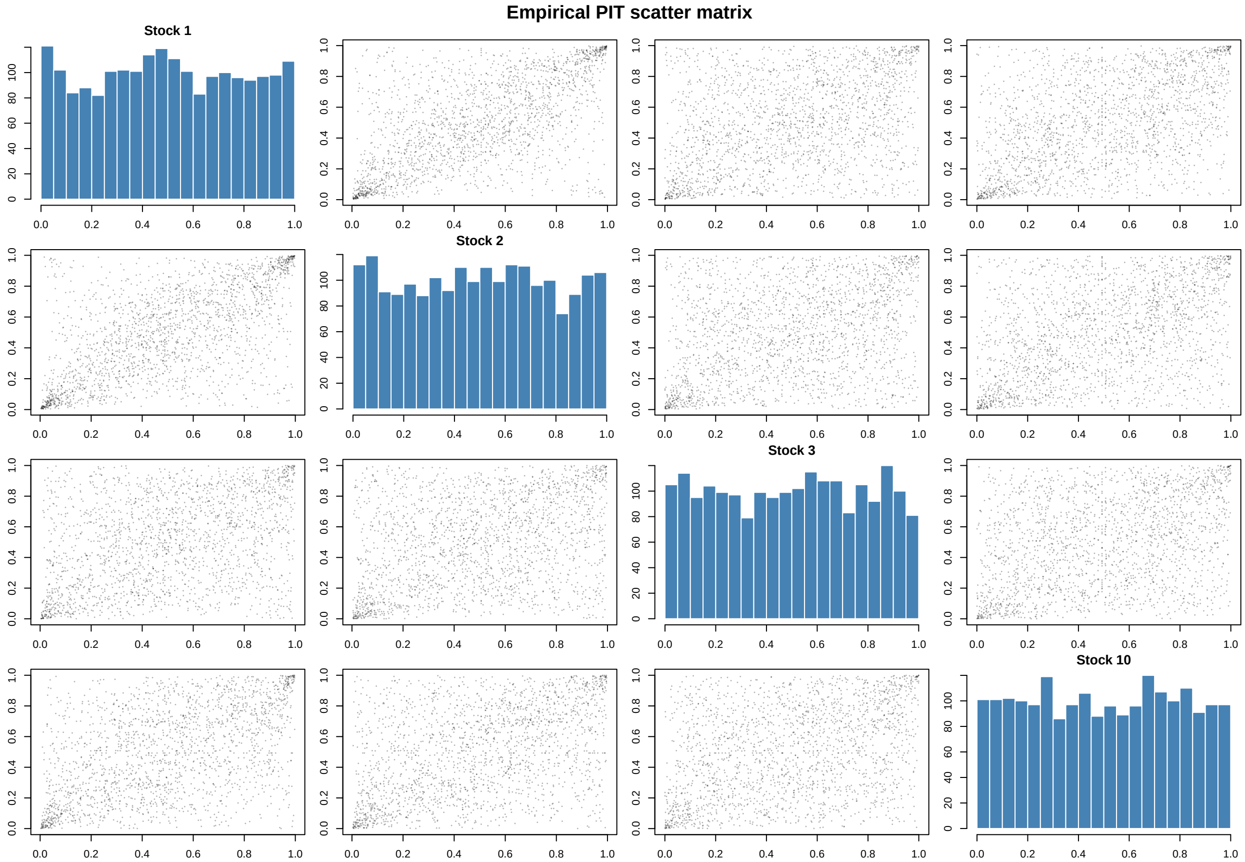
Table 3: Marginal Student-t degrees of freedom

asset	df
asset_1	1.7858
asset_2	1.3511
asset_3	3.0861
asset_10	4.0148

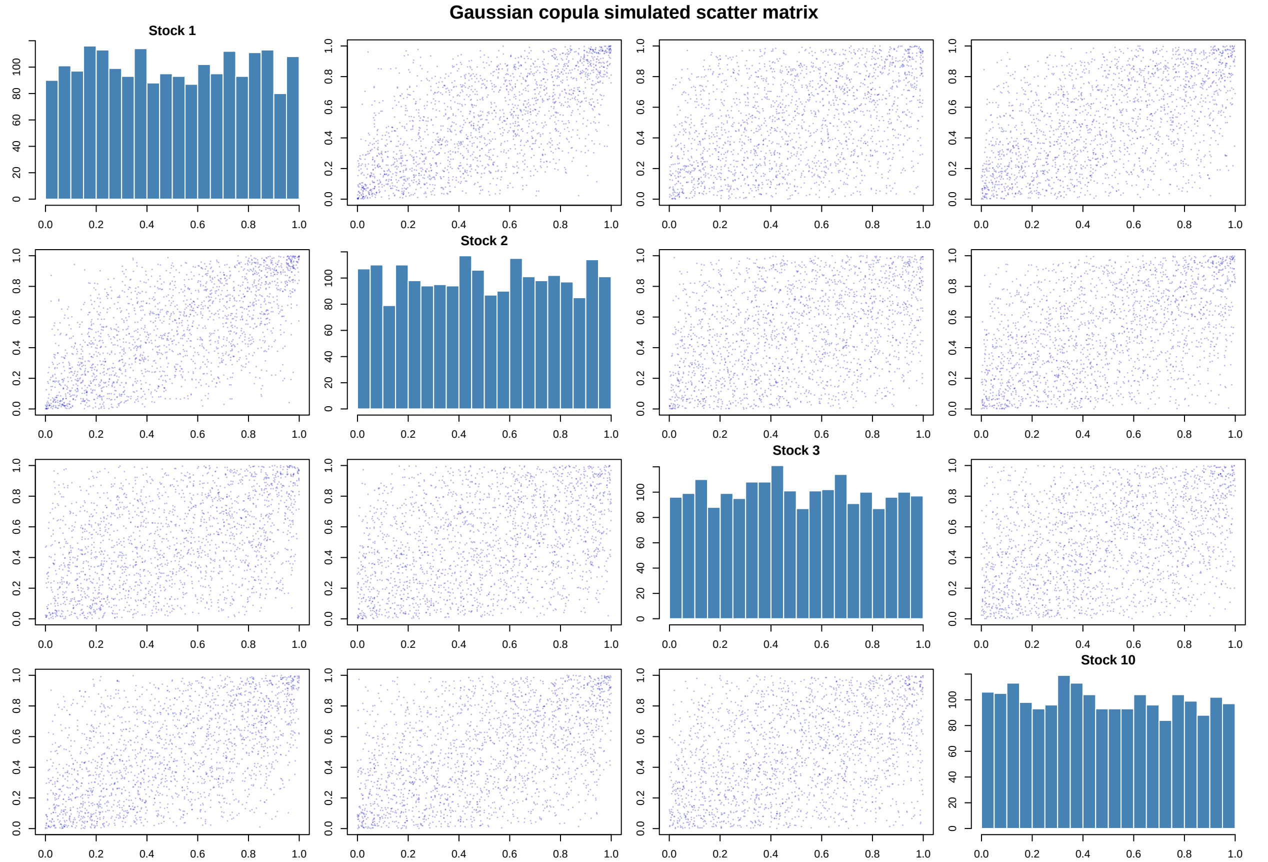
The fitted copula $\hat{\nu} = 3.76$ sits between the marginal degrees of freedom: it is higher than the fattest-tailed assets (stocks 1 and 2, with $\hat{\nu} = 1.79$ and 1.35) but lower than the lighter-tailed assets (stocks 3 and 10, with $\hat{\nu} = 3.09$ and 4.01). What matters is that the copula $\hat{\nu}$ is finite — unlike the Gaussian copula, which implicitly assumes $\nu \to \infty$ — and it is this finiteness that generates positive asymptotic tail dependence in the joint model even after the marginals have been stripped out by the PIT.

e) Simulate and compare

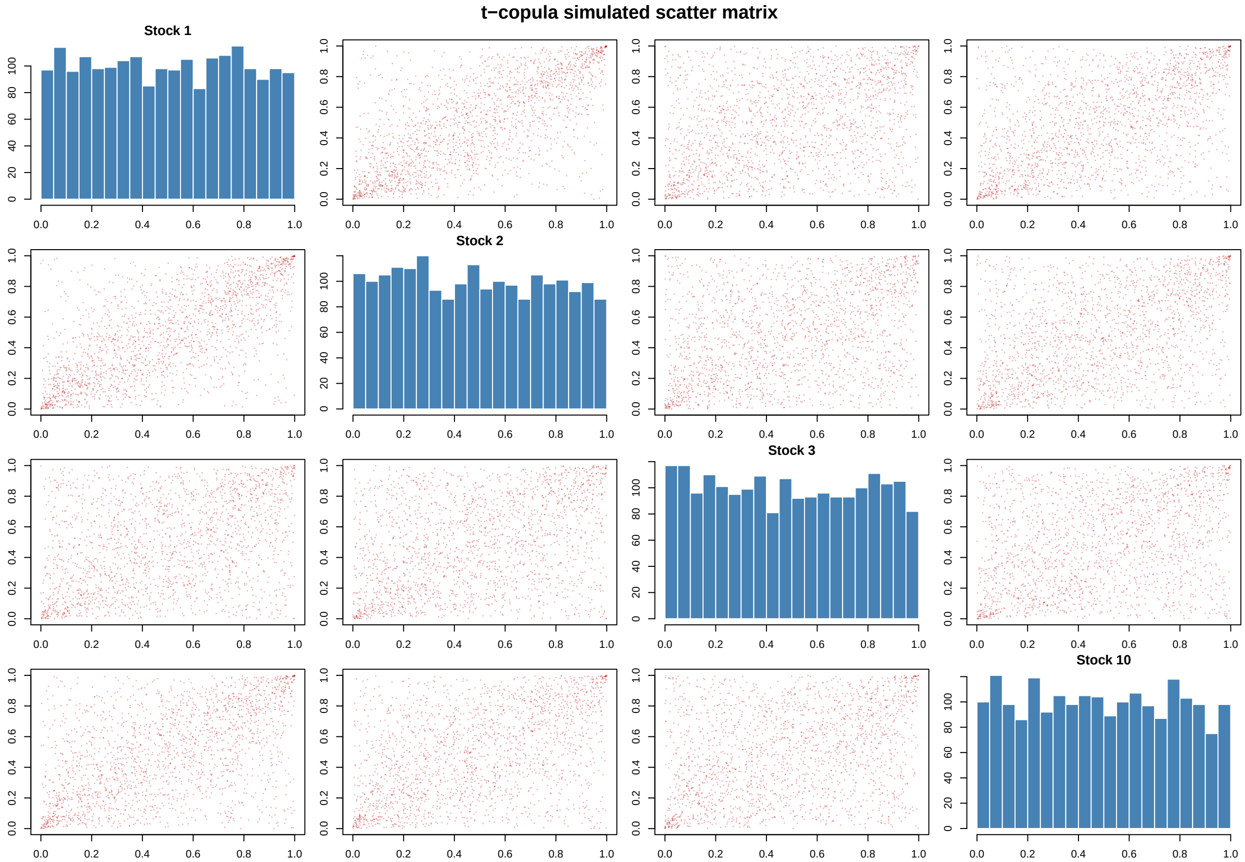
```r
set.seed(544)
n_sim <- 2000

# Simulate from Gaussian copula
u_gauss_sim <- rCopula(n_sim, gauss_fit@copula)

# Simulate from t-copula
u_t_sim     <- rCopula(n_sim, t_fit@copula)

# Helper: plot 4x4 scatter matrix
plot_pit_matrix <- function(u_mat, main_title, col_pts = rgb(0, 0, 0, 0.3)) {
  par(mfrow = c(4, 4), mar = c(2, 2, 1.5, 1), oma = c(0, 0, 2, 0))
  for (i in 1:4) {
    for (j in 1:4) {
      if (i == j) {
        hist(u_mat[, i], breaks = 20, col = "steelblue", border = "white",
             main = labels[i], xlab = "", ylab = "")
      } else {
        plot(u_mat[, j], u_mat[, i], pch = 16, cex = 0.3, col = col_pts,
             xlab = labels[j], ylab = labels[i], main = "")
      }
    }
  }
  mtext(main_title, outer = TRUE, cex = 1.2, font = 2)
}

# Empirical
plot_pit_matrix(u4_mat, "Empirical PIT scatter matrix")
```




```r
# Gaussian-simulated
plot_pit_matrix(u_gauss_sim, "Gaussian copula simulated scatter matrix",
                col_pts = rgb(0, 0, 0.7, 0.3))
```




```r
# t-simulated
plot_pit_matrix(u_t_sim, "t-copula simulated scatter matrix",
                col_pts = rgb(0.7, 0, 0, 0.3))
```



The t-copula reproduces the empirical corner mass much better than the Gaussian copula. The Gaussian copula scatter fills the interior of the unit square without concentrating in the corners, because a Gaussian copula has zero asymptotic tail dependence, no matter how correlated the margins are, extreme joint events become asymptotically independent. The t-copula, by contrast, has positive asymptotic tail dependence governed by its degrees-of-freedom parameter $\hat{\nu}$: when one asset crashes, others are more likely to crash simultaneously. This is the canonical reason finance prefers the t-copula, it captures the joint tail co-movement that actually characterizes equity portfolios during market stress, which the Gaussian copula structurally cannot.